

Lecture 17: Dataframes



INFO 1100: Introduction to Programming

learning **NumPy** gave us a very
useful tool for analyzing data

learning NumPy gave us a very
useful tool for analyzing data

but there were some restrictions...

learning NumPy gave us a very
useful tool for analyzing data

but there were some restrictions...

like what?

learning **NumPy** gave us a very
useful tool for analyzing data

but there were some restrictions...

mostly useful for
numerical data

learning **NumPy** gave us a very
useful tool for analyzing data

but there were some restrictions...

mostly useful for
numerical data

only homogeneous
data

learning **NumPy** gave us a very
useful tool for analyzing data

but there were some restrictions...

mostly useful for
numerical data

we can't label
columns

only homogeneous
data

learning NumPy gave us a very
useful tool for analyzing data

but

What data might break
these restrictions?

ns...

mostly useful for
numerical data

we can't label
columns

only homogeneous
data

the solution...

the solution...

Pandas dataframes

the solution...

Pandas dataframes

(built on top of NumPy)

Dataframes

Dataframes

What are they?

Dataframes

data tables

Dataframes

spreadsheets

Dataframes

2D NumPy arrays with
row and column labels

Dataframes

to use dataframes...

Dataframes

to use dataframes...

```
import pandas as pd
```

Dataframes

creating dataframes

Dataframes

creating dataframes
from NumPy arrays

Dataframes

creating dataframes

from NumPy arrays

```
import pandas as pd  
import numpy as np
```

Dataframes

creating dataframes

from NumPy arrays

```
import pandas as pd
import numpy as np

arr = np.random.rand(3, 2)
```

Dataframes

creating dataframes

from NumPy arrays

```
import pandas as pd
import numpy as np

arr = np.random.rand(3, 2)
df = pd.DataFrame(arr)
```

Dataframes

creating dataframes

from NumPy arrays

```
import pandas as pd
import numpy as np

arr = np.random.rand(3, 2)
df = pd.DataFrame(arr)
```


Dataframes

creating dataframes *from NumPy arrays*

```
import pandas as pd
import numpy as np

arr = np.random.rand(3, 2)
df = pd.DataFrame(arr)
```

	0	1
0	0.406	0.835
1	0.921	0.643
2	0.373	0.127

Dataframes

creating dataframes *from NumPy arrays*

```
import pandas as pd
import numpy as np

arr = np.random.rand(3, 2)
df = pd.DataFrame(arr,
    columns=['a', 'b'])
```

	a	b
0	0.406	0.835
1	0.921	0.643
2	0.373	0.127

Dataframes

creating dataframes *from NumPy arrays*

```
import pandas as pd
import numpy as np

arr = np.random.rand(3, 2)
df = pd.DataFrame(arr,
                  columns=['a', 'b'],
                  index=['I', 'II', 'III'])
```

	a	b
I	0.406	0.835
II	0.921	0.643
III	0.373	0.127

Dataframes

creating dataframes

from lists of dictionaries

Dataframes

creating dataframes

from lists of dictionaries

```
import pandas as pd
```

Dataframes

creating dataframes

from lists of dictionaries

```
import pandas as pd

l = [{ 'a': 0, 'b': 2 },
      { 'b': 1, 'c': 3 }]
```

Dataframes

creating dataframes

from lists of dictionaries

```
import pandas as pd

l = [{ 'a': 0, 'b': 2 },
      { 'b': 1, 'c': 3 }]

df = pd.DataFrame(l)
```

Dataframes

creating dataframes
from lists of dictionaries

```
import pandas as pd

l = [{ 'a': 0, 'b': 2 },
      { 'b': 1, 'c': 3 }]
df = pd.DataFrame(l)
```

	a	b	c
0	0	2	NaN
1	NaN	1	3

Dataframes

creating dataframes
from lists of dictionaries

```
import pandas as pd

l = [{ 'a': 0, 'b': 2 },
      { 'b': 1, 'c': 3 }]
df = pd.DataFrame(l)
```

	a	b	c
0	0	2	NaN
1	NaN	1	3

Dataframes

creating dataframes
from lists of dictionaries

```
import pandas as pd

l = [{ 'a': 0, 'b': 2 },
      { 'b': 1, 'c': 3 }]
df = pd.DataFrame(l)
```

	a	b	c
0	0	2	NaN
1	NaN	1	3

Dataframes

creating dataframes
from lists of dictionaries

```
import pandas as pd

l = [{ 'a': 0, 'b': 2 },
      { 'b': 1, 'c': 3 }]
df = pd.DataFrame(l)
```

missing values

	a	b	c
0	0	2	NaN
1	NaN	1	3

Dataframes

creating dataframes

from a dictionary with list values

Dataframes

creating dataframes

from a dictionary with list values

```
import pandas as pd

d = { 'name': [ 'Jane', 'Tom',
               'Lisa'], 'num': [2, 4, 3] }
df = pd.DataFrame(d)
```

	name	num
0	Jane	2
1	Tom	4
2	Lisa	3

Dataframes

creating dataframes

from a dictionary with list values

```
import pandas as pd
```

```
d = { 'name': ['Jane', 'Tom',  
             'Lisa'], 'num': [2, 4, 3] }
```

```
df = pd.DataFrame(d)
```

	name	num
0	Jane	2
1	Tom	4
2	Lisa	3

Dataframes

creating dataframes

from a file

Dataframes

creating dataframes
from a file

probably the most useful!

Dataframes

creating dataframes

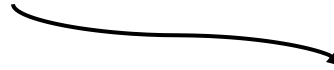
from a file

often csv files

Dataframes

creating dataframes
from a file

often csv files



comma-separated-values

Dataframes

creating dataframes

from a file

often csv files

*commonly used to store tabular
data of all types*

Dataframes

creating dataframes

from a file

often csv files

*commonly used to store tabular
data of all types*

**your final project data is probably in this
format!**

Dataframes

creating dataframes
from a file

books.csv

title	# words
1984	80000
Fahrenheit 451	46000
Catch-22	174000
The Three-Body Problem	104000

Dataframes

creating dataframes
from a file

```
import pandas as pd
```

Dataframes

creating dataframes

from a file

```
import pandas as pd
```

```
df = pd.read_csv("books.csv")
```

Dataframes

creating dataframes *from a file*

```
import pandas as pd
```

```
df = pd.read_csv("books.csv")
```

	title	# words
0	1984	80000
1	Fahrenheit 451	46000
2	Catch-22	174000
3	The Three-Body Problem	104000

Dataframes

exploring dataframes

Dataframes

exploring dataframes

```
import pandas as pd
```

```
df = pd.read_csv("books.csv")
```

	title	# words
0	1984	80000
1	Fahrenheit 451	46000
2	Catch-22	174000
3	The Three-Body Problem	104000

Dataframes

exploring dataframes

```
import pandas as pd

df = pd.read_csv("books.csv")
print(df.head())
```

	title	# words
0	1984	80000
1	Fahrenheit 451	46000
2	Catch-22	174000
3	The Three-Body Problem	104000

Dataframes

exploring dataframes

```
import pandas as pd

df = pd.read_csv("books.csv")
print(df.head())
```

**shows the first five
rows**



	title	# words
0	1984	80000
1	Fahrenheit 451	46000
2	Catch-22	174000
3	The Three-Body Problem	104000

Dataframes

exploring dataframes

```
import pandas as pd

df = pd.read_csv("books.csv")
print(df.head())
print(len(df))
```

	title	# words
0	1984	80000
1	Fahrenheit 451	46000
2	Catch-22	174000
3	The Three-Body Problem	104000

Dataframes

exploring dataframes

```
import pandas as pd

df = pd.read_csv("books.csv")
print(df.head())
print(len(df)) → 4
```

	title	# words
0	1984	80000
1	Fahrenheit 451	46000
2	Catch-22	174000
3	The Three-Body Problem	104000

Dataframes

exploring dataframes

```
import pandas as pd

df = pd.read_csv("books.csv")
print(df.head())
print(len(df))
print(df.shape)
```

	title	# words
0	1984	80000
1	Fahrenheit 451	46000
2	Catch-22	174000
3	The Three-Body Problem	104000

Dataframes

exploring dataframes

```
import pandas as pd

df = pd.read_csv("books.csv")
print(df.head())
print(len(df))
print(df.shape) → (4, 2)
```

	title	# words
0	1984	80000
1	Fahrenheit 451	46000
2	Catch-22	174000
3	The Three-Body Problem	104000

Dataframes

exploring dataframes

```
import pandas as pd

df = pd.read_csv("books.csv")
print(df.head())
print(len(df))
print(df.shape)
print(df.columns)
```

	title	# words
0	1984	80000
1	Fahrenheit 451	46000
2	Catch-22	174000
3	The Three-Body Problem	104000

Dataframes

exploring dataframes

```
import pandas as pd

df = pd.read_csv("books.csv")
print(df.head())
print(len(df))
print(df.shape)
print(df.columns)
```

`Index(['title', '# words'],
 dtype='object')`



	title	# words
0	1984	80000
1	Fahrenheit 451	46000
2	Catch-22	174000
3	The Three-Body Problem	104000

Dataframes

exploring dataframes

```
import pandas as pd

df = pd.read_csv("books.csv")
print(df.head())
print(len(df))
print(df.shape)
print(df.columns)
print(df.index)
```

	title	# words
0	1984	80000
1	Fahrenheit 451	46000
2	Catch-22	174000
3	The Three-Body Problem	104000

Dataframes

exploring dataframes

```
import pandas as pd
```

```
df = pd.read_csv("books.csv")
```

```
print(df.head())
```

```
print(len(df))
```

```
print(df.shape)
```

```
print(df.columns)
```

```
print(df.index)
```

RangeIndex(start=0, stop=4, step=1)

	title	# words
0	1984	80000
1	Fahrenheit 451	46000
2	Catch-22	174000
3	The Three-Body Problem	104000

Dataframes

indexing dataframes

Dataframes

indexing dataframes

```
import pandas as pd
```

```
df = pd.read_csv("books.csv")
```

	title	# words
0	1984	80000
1	Fahrenheit 451	46000
2	Catch-22	174000
3	The Three-Body Problem	104000

Dataframes

indexing dataframes

```
import pandas as pd

df = pd.read_csv("books.csv")
print(df["# words"])
```

	title	# words
0	1984	80000
1	Fahrenheit 451	46000
2	Catch-22	174000
3	The Three-Body Problem	104000

Dataframes

indexing dataframes

```
import pandas as pd
```

```
df = pd.read_csv("books.csv")
```

```
print(df["# words"])
```

0	8000
1	46000
2	174000
3	104000

→ **Name: # words, dtype: int64**

	title	# words
0	1984	80000
1	Fahrenheit 451	46000
2	Catch-22	174000
3	The Three-Body Problem	104000

Dataframes

indexing dataframes

```
import pandas as pd
```

```
df = pd.read_csv("books.csv")
```

```
print(df["# words"])
```

0	8000
1	46000
2	174000
3	104000

Series

Name: # words, dtype: int64

	title	# words
0	1984	80000
1	Fahrenheit 451	46000
2	Catch-22	174000
3	The Three-Body Problem	104000

Dataframes

indexing dataframes

```
import pandas as pd

df = pd.read_csv("books.csv")
print(df["# words"])
print(df.loc[0])
```

	title	# words
0	1984	80000
1	Fahrenheit 451	46000
2	Catch-22	174000
3	The Three-Body Problem	104000

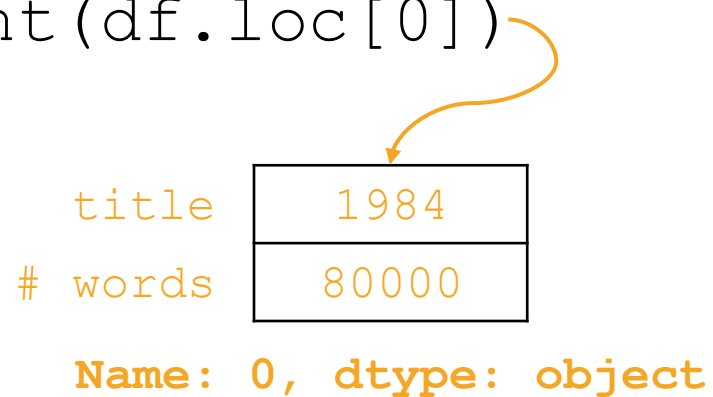
Dataframes

indexing dataframes

```
import pandas as pd

df = pd.read_csv("books.csv")
print(df["# words"])
print(df.loc[0])
```

title 1984
words 80000
Name: 0, dtype: object



	title	# words
0	1984	80000
1	Fahrenheit 451	46000
2	Catch-22	174000
3	The Three-Body Problem	104000

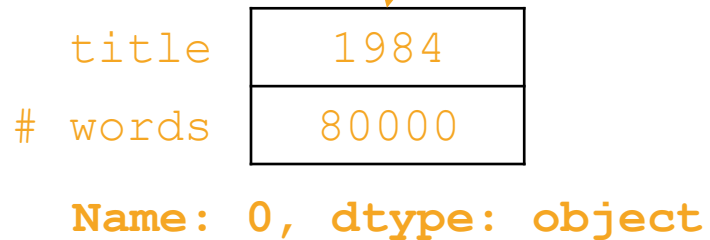
Dataframes

indexing dataframes

```
import pandas as pd

df = pd.read_csv("books.csv")
print(df["# words"])
print(df.loc[0])
```

title 1984
words 80000
Name: 0, dtype: object



**looks at the
index label**

	title	# words
0	1984	80000
1	Fahrenheit 451	46000
2	Catch-22	174000
3	The Three-Body Problem	104000

Dataframes

indexing dataframes

```
import pandas as pd

df = pd.read_csv("books.csv")
print(df["# words"])
print(df.loc[0])
print(df.iloc[0])
```

	title	# words
0	1984	80000
1	Fahrenheit 451	46000
2	Catch-22	174000
3	The Three-Body Problem	104000

Dataframes

indexing dataframes

```
import pandas as pd
```

```
df = pd.read_csv("books.csv")
```

```
print(df["# words"])
```

```
print(df.loc[0])
```

```
print(df.iloc[0])
```

title 1984
words 80000
Name: 0, dtype: object

	title	# words
0	1984	80000
1	Fahrenheit 451	46000
2	Catch-22	174000
3	The Three-Body Problem	104000

Dataframes

indexing dataframes

```
import pandas as pd

df = pd.read_csv("books.csv")
print(df["# words"])
print(df.loc[0])
print(df.iloc[0])
```

title 1984
words 80000
Name: 0, dtype: object

looks at the
integer position

	title	# words
0	1984	80000
1	Fahrenheit 451	46000
2	Catch-22	174000
3	The Three-Body Problem	104000

Dataframes

filtering dataframes

Dataframes

filtering dataframes

```
import pandas as pd
```

```
df = pd.read_csv("books.csv")
```

	title	# words
0	1984	80000
1	Fahrenheit 451	46000
2	Catch-22	174000
3	The Three-Body Problem	104000

Dataframes

filtering dataframes

```
import pandas as pd

df = pd.read_csv("books.csv")
print(df[df["# words"] > 100000])
```

	title	# words
0	1984	80000
1	Fahrenheit 451	46000
2	Catch-22	174000
3	The Three-Body Problem	104000

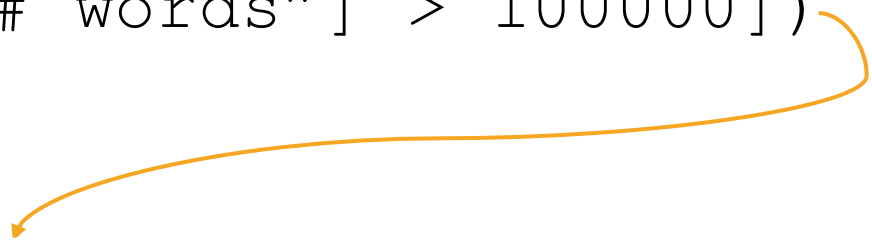
Dataframes

filtering dataframes

```
import pandas as pd
```

```
df = pd.read_csv("books.csv")
```

```
print(df[df["# words"] > 100000])
```



	title	# words
2	Catch-22	174000
3	The Three-Body Problem	104000

	title	# words
0	1984	80000
1	Fahrenheit 451	46000
2	Catch-22	174000
3	The Three-Body Problem	104000

Dataframes

filtering dataframes

```
import pandas as pd
```

```
df = pd.read_csv("books.csv")
```

```
print(df[df["# words"] > 100000])
```

```
print(df[df["title"].str.contains("22")])
```

	title	# words
0	1984	80000
1	Fahrenheit 451	46000
2	Catch-22	174000
3	The Three-Body Problem	104000

Dataframes

filtering dataframes

```
import pandas as pd
```

```
df = pd.read_csv("books.csv")
```

```
print(df[df["# words"] > 100000])
```

```
print(df[df["title"].str.contains("22")])
```

	title	# words
2	Catch-22	174000

	title	# words
0	1984	80000
1	Fahrenheit 451	46000
2	Catch-22	174000
3	The Three-Body Problem	104000

Dataframes

analyzing dataframes

Dataframes

analyzing dataframes

```
import pandas as pd
```

```
df = pd.read_csv("books.csv")
```

	title	# words
0	1984	80000
1	Fahrenheit 451	46000
2	Catch-22	174000
3	The Three-Body Problem	104000

Dataframes

analyzing dataframes

```
import pandas as pd

df = pd.read_csv("books.csv")
print(df["# words"].max())
```

	title	# words
0	1984	80000
1	Fahrenheit 451	46000
2	Catch-22	174000
3	The Three-Body Problem	104000

Dataframes

analyzing dataframes

```
import pandas as pd
```

```
df = pd.read_csv("books.csv")
```

```
print(df["# words"].max()) → 174000
```

	title	# words
0	1984	80000
1	Fahrenheit 451	46000
2	Catch-22	174000
3	The Three-Body Problem	104000

Dataframes

analyzing dataframes

```
import pandas as pd

df = pd.read_csv("books.csv")
print(df["# words"].max())
print(df["# words"].mean())
```

	title	# words
0	1984	80000
1	Fahrenheit 451	46000
2	Catch-22	174000
3	The Three-Body Problem	104000

Dataframes

analyzing dataframes

```
import pandas as pd
```

```
df = pd.read_csv("books.csv")
```

```
print(df["# words"].max())
```

```
print(df["# words"].mean()) → 101000.0
```

	title	# words
0	1984	80000
1	Fahrenheit 451	46000
2	Catch-22	174000
3	The Three-Body Problem	104000

Dataframes

analyzing dataframes

```
import pandas as pd
```

```
df = pd.read_csv("books.csv")
```

**let's add a
column!**

	title	# words
0	1984	80000
1	Fahrenheit 451	46000
2	Catch-22	174000
3	The Three-Body Problem	104000

Dataframes

analyzing dataframes

```
import pandas as pd
```

```
df = pd.read_csv("books.csv")
```

```
df["genre"] = ["classics", "classics", "classics", "sci-fi"]
```

	title	# words
0	1984	80000
1	Fahrenheit 451	46000
2	Catch-22	174000
3	The Three-Body Problem	104000

Dataframes

analyzing dataframes

```
import pandas as pd
```

```
df = pd.read_csv("books.csv")
```

```
df["genre"] = ["classics", "classics", "classics", "sci-fi"]
```

	title	# words	genre
0	1984	80000	classics
1	Fahrenheit 451	46000	classics
2	Catch-22	174000	classics
3	The Three-Body Problem	104000	sci-fi

Dataframes

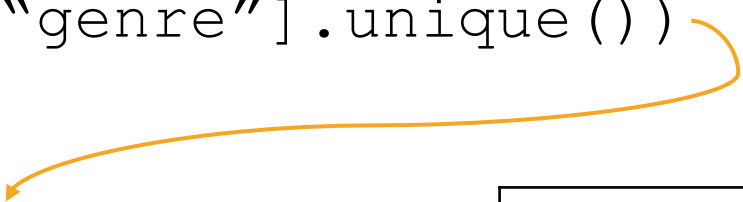
analyzing dataframes

```
import pandas as pd
```

```
df = pd.read_csv("books.csv")
```

```
df["genre"] = ["classics", "classics", "classics", "sci-fi"]
```

```
print(df["genre"].unique())
```



```
array(['classics',  
      'sci-fi'],  
      dtype=object)
```

	title	# words	genre
0	1984	80000	classics
1	Fahrenheit 451	46000	classics
2	Catch-22	174000	classics
3	The Three-Body Problem	104000	sci-fi

Dataframes

analyzing dataframes

so much more!

Dataframes

what else do you want
to do with dataframes?

Dataframes

what else do you want
to do with dataframes?

for your final project?

Example 1

```
import pandas as pd

df = pd.DataFrame({
    "name": ["Ana", "Ben", "Cara", "Dan", "Eve", "Finn"],
    "age": [25, 40, 30, 25, 35, 40],
    "city": ["NYC", "LA", "NYC", "LA", "NYC", "NYC"],
    "visits": [6, 2, 8, 1, 4, 6]
})

nyc = df[df["city"] == "NYC"]
print(len(nyc))
print(nyc["visits"].sum())
print(nyc["visits"].mean())
```

Example 2

```
import pandas as pd

df = pd.DataFrame({
    "name": ["Ana", "Ben", "Cara", "Dan", "Eve", "Finn"],
    "age": [25, 40, 30, 25, 35, 40],
    "city": ["NYC", "LA", "NYC", "LA", "NYC", "NYC"],
    "visits": [6, 2, 8, 1, 4, 6]
})

avg = df["age"].mean()
above = df[df["age"] > avg]
print(avg, len(above))
print(above["name"].tolist())
```

Example 3

```
import pandas as pd

df = pd.DataFrame({
    "name": ["Ana", "Ben", "Cara", "Dan", "Eve", "Finn"],
    "age": [25, 40, 30, 25, 35, 40],
    "city": ["NYC", "LA", "NYC", "LA", "NYC", "NYC"],
    "visits": [6, 2, 8, 1, 4, 6]
})

df["adult"] = df["age"] >= 35
print(df["adult"].sum())
print(df[df["adult"]]["name"].tolist())
```

Example 4

```
import pandas as pd

df = pd.DataFrame({
    "name": ["Ana", "Ben", "Cara", "Dan", "Eve", "Finn"],
    "age": [25, 40, 30, 25, 35, 40],
    "city": ["NYC", "LA", "NYC", "LA", "NYC", "NYC"],
    "visits": [6, 2, 8, 1, 4, 6]
})

frequent = df[df["visits"] >= 5]
print(len(frequent))
print(frequent["age"].mean())
```

Questions?

(... I'll wait for 2)

For tomorrow:

(07/29 @ 11:30am)

- Lab 17
- Readings
Look at Data
- GRQs